

Monster Game II

Time limit: 1.00 s

Memory limit: 512 MB

You are playing a game that consists of n levels. Each level has a monster. On levels $1, 2, \dots, n - 1$, you can either kill or escape the monster. However, on level n you must kill the final monster to win the game.

Killing a monster takes sf time where s is the monster's strength and f is your skill factor. After killing a monster, you get a new skill factor (lower skill factor is better). What is the minimum total time in which you can win the game?

Input

The first input line has two integers n and x : the number of levels and your initial skill factor.

The second line has n integers $s_1, s_2, \dots, s_n$: each monster's strength.

The third line has n integers $f_1, f_2, \dots, f_n$: your new skill factor after killing a monster.

Output

Print one integer: the minimum total time to win the game.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x \leq 10^6$
- $1 \leq s_i, f_i \leq 10^6$

Example

Input	Output
5 100 50 20 30 90 30 60 20 20 10 90	2600

Explanation: The best way to play is to kill the second and fifth monster.